

Project Title

**Implementation of an Open-Source Security Operations Center
Monitoring Solution Using Wazuh and Suricata**

Project Proposal

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1. Introduction

In the contemporary digital landscape, organizations face an unprecedented proliferation of sophisticated cyber threats that pose significant risks to information security infrastructure. Security Operations Centers (SOCs) have emerged as critical components of organizational cybersecurity strategy, serving as centralized units responsible for continuous monitoring, detection, analysis, and response to security incidents. SOC analysts leverage diverse security tools and technologies to maintain comprehensive visibility across endpoint systems and network infrastructure, enabling real-time threat detection and incident response capabilities.

The adoption of open-source security solutions has gained considerable traction as viable alternatives to proprietary commercial platforms, particularly within academic institutions and small-to-medium enterprises where budgetary constraints may limit access to enterprise-grade solutions. This proposal presents a comprehensive framework for implementing a functional SOC monitoring environment utilizing Wazuh, an open-source Security Information and Event Management (SIEM) platform with Host-based Intrusion Detection System (HIDS) capabilities, integrated with Suricata, a high-performance Network Intrusion Detection System (NIDS). The proposed implementation demonstrates the feasibility of deploying enterprise-level security monitoring capabilities within resource-constrained environments while maintaining professional operational standards.

2. Problem Statement

Security analysts in modern SOC environments confront the challenge of processing and analyzing substantial volumes of security events generated across distributed endpoint systems and network infrastructure. The absence of centralized monitoring and event correlation mechanisms significantly impedes the timely detection of malicious activities, including but not limited to brute-force authentication attempts, malware infections, unauthorized access attempts, and anomalous network traffic patterns. Furthermore, numerous educational institutions and organizations encounter substantial barriers to implementing commercial SOC platforms due to prohibitive licensing costs, technical complexity, and resource requirements.

This research initiative addresses the critical need for a cost-effective, pedagogically valuable, and operationally practical SOC monitoring solution that can be successfully implemented, configured, and

demonstrated within a constrained timeframe of four weeks. The proposed solution aims to bridge the gap between theoretical cybersecurity knowledge and practical security operations skills while demonstrating the viability of open-source security tools in real-world monitoring scenarios.

3. Research Aim and Objectives

3.1 Aim

The primary aim of this research project is to design, implement, and evaluate a scalable SOC monitoring environment utilizing open-source security tools to effectively detect, analyze, and document security threats within a controlled network environment. This implementation will demonstrate the practical applicability of open-source security solutions in establishing robust security monitoring capabilities comparable to commercial alternatives.

3.2 Objectives

The specific objectives of this research project are:

1. To deploy and configure Wazuh as a centralized SIEM platform with integrated host-based intrusion detection capabilities for comprehensive security event aggregation and analysis
2. To implement Suricata as a network intrusion detection system for real-time network traffic monitoring and threat identification
3. To establish seamless integration between Suricata and Wazuh for unified security event correlation and centralized alert management
4. To implement built-in security rules and detection mechanisms aligned with contemporary threat patterns and attack methodologies
5. To conduct controlled simulation of common attack vectors
6. To demonstrate practical SOC analyst workflows including alert triage, incident investigation, and security event analysis through the integrated monitoring platform

4. Technology Overview and Justification

4.1 Wazuh Security Platform

Wazuh originated in 2015 as a fork of the OSSEC open-source HIDS project, created and maintained by Wazuh, Inc., a cybersecurity company headquartered in San Jose, California. The project is sponsored by Wazuh, Inc. under an open-core model, releasing the core platform as free software under the GNU

General Public License (GPL) while providing optional commercial support. The platform delivers log data analysis, file integrity monitoring, vulnerability detection, configuration assessment, incident response, and regulatory compliance monitoring.

4.2 Suricata Network Intrusion Detection System

Suricata was created in 2009 by Victor Julien and officially released in 2010 by the Open Information Security Foundation (OISF), a non-profit consortium originally funded by the U.S. Department of Homeland Security (DHS). OISF serves as the primary sponsor of Suricata, governing its development as a community-owned, vendor-neutral network security engine with support from consortium members including Cisco, Fortinet, and various national cybersecurity agencies. The system utilizes a multi-threaded architecture to achieve high-throughput network monitoring while supporting deep packet inspection, protocol identification, file extraction, and TLS certificate logging.

5. Methodology and Implementation Framework

This research project employs a practical implementation methodology conducted within a controlled virtualized laboratory environment. The experimental infrastructure will be constructed using enterprise virtualization platforms to ensure reproducibility and isolation from production networks. The implementation follows a systematic, phased approach designed to ensure proper integration, testing, and validation of all system components.

5.1 Laboratory Infrastructure Design

The laboratory environment simulates a small enterprise SOC scenario in which a security analyst monitors a local network for threats using open-source tools. The setup consists of five machines:

- **Wazuh Server:** runs the Wazuh Manager, Indexer, and Dashboard
- **Suricata Sensor:** dedicated network monitoring node running Suricata in IDS mode, captures traffic on the internal interface and forwards EVE JSON alerts to the Wazuh Manager
- **Linux Endpoint:** has the Wazuh agent installed for host-based monitoring
- **Windows Endpoint:** has the Wazuh agent installed for Windows event log collection
- **Attacker Machine:** used to simulate controlled attack scenarios such as brute-force, port scanning, and web application attacks

5.2 Implementation Phases

The implementation process is structured into distinct phases:

Phase 1: Foundation Infrastructure Deployment

This initial phase encompasses virtual machine provisioning, network configuration, and Wazuh server installation. Activities include operating system hardening, installation of the Wazuh manager and indexer components, configuration of the Wazuh dashboard interface, and validation of core system functionality through initial testing procedures.

Phase 2: Agent Deployment and Module Configuration

The second phase focuses on deploying Wazuh agents across monitored endpoints, establishing secure communication channels with the central manager, and activating baseline module and rule configurations. The following modules and their corresponding alert rules are configured:

- File Integrity Monitoring (FIM): alert rules 550–552
- Authentication Monitor: alert rules 5710, 5712 (SSH) and rule 60122 (Windows Event ID 4625)
- Sysmon Integration: alert rules 61603, 92200
- Malware Detection: alert rule 87105
- Log Analysis for Outbound Connections: Wazuh rule 533 and Suricata ET POLICY rules
- Suricata ET SCAN ruleset: sid:2009582, sid:2010935
- Suricata ET WEB_SERVER ruleset: sid:2006446
- Suricata ET SCAN and ET DOS ruleset: sid:2100469, sid:2200074

Alert threshold tuning is applied across all modules to reduce false positives while maintaining effective detection coverage across all monitored endpoints.

Phase 3: Network Monitoring Integration

This phase involves Suricata installation and configuration for network traffic capture and analysis. Key activities include deployment of Suricata on the network monitoring node, configuration of network interfaces for traffic mirroring, integration of Emerging Threats and community rulesets, optimization of rule performance, and establishment of log forwarding mechanisms to the Wazuh platform for consolidated event analysis.

Phase 4: Testing, Validation, and Documentation

The final phase encompasses comprehensive system testing through controlled attack simulations, validation of detection capabilities, and documentation of findings. The following twelve test cases,

grouped into five categories, were selected for being practical to simulate within the lab environment and directly relevant to the tools deployed.

Authentication and Access Attacks: (1) SSH brute-force attack using Hydra against a Linux target, verifying Wazuh rule 5710/5712 alerting on repeated failed login attempts; (2) RDP brute-force attack using Hydra against a Windows target, validating Windows Event ID 4625 detection and alert generation.

Network Reconnaissance and Scanning: (3) TCP SYN port scan using Nmap against multiple hosts, triggering Suricata ET SCAN rules for host discovery (sid:2009582); (4) Service version enumeration scan using Nmap -sV to detect banner grabbing behavior, validated against Suricata ET SCAN protocol-level detection rules (sid:2010935).

File Integrity and System Tampering: (5) Unauthorized modification of /etc/passwd on a Linux endpoint to trigger Wazuh File Integrity Monitoring alerts; (6) Web shell upload to a monitored web directory to simulate server compromise and validate FIM detection on new file creation; (7) Addition of a malicious cron job entry on Linux to test persistence mechanism detection.

Malware Simulation and Process Execution: (8) EICAR test file execution to validate malware detection using Wazuh built-in rule 87105 without using real malware; (9) Suspicious PowerShell execution on Windows using an encoded command to trigger Wazuh Sysmon rules 61603 and 92200 for obfuscated script detection; (10) Netcat listener establishment to simulate a Command and Control (C2) communication channel, validated against Suricata ET POLICY rules and Wazuh rule 533 for outbound connection alerting.

Network Protocol and Injection Attacks: (11) SQL injection attempt against a DVWA (Damn Vulnerable Web Application) instance to test Suricata ET WEB_SERVER ruleset detection (sid:2006446); (12) ICMP flood and ping sweep to validate volumetric traffic detection using Suricata ET SCAN and ET DOS rules (sid:2100469, sid:2200074).

Each test case will be documented with the attack tool and command used, the expected Wazuh or Suricata rule triggered, the actual alert observed in the dashboard, and the recommended SOC analyst response action.

5.3 Log Types and Implementation Levels

The implementation collects and processes specific log types selected based on the attack scenarios being tested.

Log Type	Source	Attack Scenario Detected
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Authentication Log	/var/log/auth.log	SSH brute-force, RDP brute-force, unauthorized login after attack
Audit Log	auditd	File tampering, privilege escalation, suspicious system calls
Windows Event Log	Security Channel (Event ID 4624/4625)	RDP brute-force — failed and successful logon attempts
Windows Event Log	Security Channel (Event ID 4688)	PowerShell obfuscated execution, Meterpreter process spawning
Windows Event Log	Security Channel (Event ID 4720/4726)	Account creation/deletion during post-exploitation
File Integrity Monitoring (FIM)	/etc, /bin (Linux); C:\Windows\System32 (Windows)	Web shell upload, cron persistence, registry modification, critical file deletion

Implementation Levels

The system operates at three levels. At the **collection level**, all logs are ingested raw into the Wazuh indexer for storage and forensic review. At the **detection level**, Wazuh’s rule engine analyzes logs in real time and assigns each alert a severity on a scale of 0–15: levels 0–6 are informational or debug events requiring no action; levels 7–9 indicate low-severity events such as a single failed login; levels 10–12 indicate medium-severity suspicious activity such as repeated failures or scan patterns; levels 13–15 indicate high-severity confirmed attacks such as successful intrusion or critical file modification. Suricata uses a separate priority scale of 1–3 (1 = high, 2 = medium, 3 = low), which will be mapped to the equivalent Wazuh severity range during ingestion. At the **correlation level**, Wazuh links alerts from both Suricata and its agents to detect multi-stage attack chains — for example, correlating a Suricata port scan alert with a subsequent successful SSH login on the same host to flag a likely intrusion sequence.

6. Implementation Timeline and Milestones

The project implementation follows a structured four-week timeline designed to ensure systematic development, testing, and validation of all system components. The following table delineates the planned activities and deliverables for each phase:

Phase/Week	Activities and Deliverables
Week 1	Infrastructure Foundation: Environment setup; Wazuh Server installation; Network connectivity validation and baseline security hardening
Week 2	Agent Deployment and Module Configuration: Wazuh agent installation on Linux Endpoint and Windows Endpoint, followed by agent-manager communication establishment; Module configuration and alert threshold tuning applied across all active modules.
Week 3	Network Monitoring Integration: Suricata installation; Integration of Emerging Threats ruleset covering ET SCAN, ET DOS, ET POLICY, and ET WEB_SERVER categories; Suricata EVE JSON log forwarding to the Wazuh Manager configured, followed by unified event correlation testing and validation.
Week 4	Validation and Documentation: Controlled attack scenarios; Alert generation verification and analysis workflow documentation; SOC analyst operational procedure demonstration; Comprehensive technical documentation preparation; Final system demonstration and presentation delivery

7. Expected Outcomes and Deliverables

Upon successful completion of this implementation project, the following outcomes and deliverables are anticipated:

- A fully functional SOC monitoring environment demonstrating integrated host-based and network-based threat detection capabilities
- Documented configuration procedures and best practices for deploying Wazuh and Suricata in educational or small enterprise environments
- Validated detection capabilities for common attack vectors including authentication attacks, network reconnaissance, and system compromise indicators
- Demonstration of practical SOC analyst workflows including alert triage, incident investigation, and security event correlation
- Comprehensive technical documentation suitable for knowledge transfer and educational purposes

8. Significance and Contributions

This research project delivers significant value across multiple dimensions of cybersecurity education and practical security operations. From a pedagogical perspective, the implementation provides hands-on experience with enterprise-grade security monitoring tools that directly align with industry-standard SOC analyst competencies. Students and security practitioners gain practical exposure to real-world security tools, detection methodologies, and incident analysis procedures that are directly transferable to professional security operations environments.

The project demonstrates the viability of open-source security solutions as legitimate alternatives to commercial SIEM platforms, particularly relevant for organizations operating under budgetary constraints. By documenting the implementation process, configuration requirements, and operational capabilities, this work contributes to the broader cybersecurity community's understanding of practical open-source security tool deployment. The accelerated four-week implementation timeline further validates the accessibility of these technologies for rapid deployment scenarios, emergency response situations, or time-constrained educational settings.

Additionally, the integrated approach combining host-based and network-based monitoring exemplifies security best practices that emphasize defense-in-depth strategies. This holistic monitoring capability provides comprehensive visibility across multiple attack surfaces, demonstrating how complementary security tools can be orchestrated to achieve enhanced threat detection capabilities beyond what individual tools provide in isolation.

9. Scope and Limitations

9.1 Project Scope

This implementation project focuses specifically on establishing core SOC monitoring capabilities within a controlled laboratory environment. The scope encompasses deployment of centralized security monitoring infrastructure, configuration of threat detection mechanisms for common attack patterns, and demonstration of fundamental SOC analyst operational procedures. The project prioritizes practical implementation experience and validation of detection capabilities over comprehensive coverage of all possible attack scenarios or advanced security automation features.

9.2 Acknowledged Limitations

Several limitations are acknowledged due to time and resource constraints inherent in the four-week implementation timeline. The laboratory environment operates at limited scale compared to enterprise deployments, restricting the evaluation of performance characteristics under high-volume event conditions. Advanced security orchestration, automation, and response (SOAR) capabilities, while valuable in production environments, are excluded from the current scope to maintain project feasibility within the allocated timeframe.

The attack simulation scenarios, while representative of common threat patterns, do not encompass the full spectrum of sophisticated attack techniques employed by advanced persistent threat actors. Enterprise-scale considerations such as high-availability architecture, distributed deployment models, long-term log retention strategies, and regulatory compliance frameworks are not addressed within this implementation. The project recognizes these limitations as opportunities for future research and extended implementation efforts beyond the current timeline constraints.

10. Conclusion

This proposal presents a comprehensive framework for implementing a functional Security Operations Center monitoring environment utilizing the open-source security platforms Wazuh and Suricata. The proposed four-week implementation timeline demonstrates the accessibility and practicality of deploying enterprise-grade security monitoring capabilities within resource-constrained environments, making advanced security operations knowledge and tools available to educational institutions and small-to-medium organizations.

The integration of host-based and network-based monitoring capabilities provides comprehensive threat detection coverage that aligns with contemporary security best practices and industry-standard SOC operations. Through systematic deployment, configuration, and validation of these security tools, the project delivers practical experience with real-world security monitoring workflows while contributing to the broader understanding of open-source security solution viability.

By successfully completing this implementation, participants will develop essential SOC analyst competencies, gain hands-on experience with industry-relevant security tools, and demonstrate the effective application of open-source technologies in addressing contemporary cybersecurity challenges. The deliverables from this project will serve as valuable educational resources and practical reference materials for future security monitoring implementations.